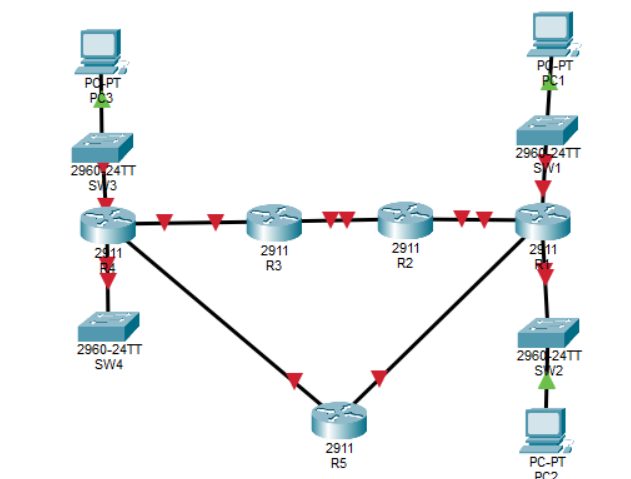


TOPOLOGY



R1 Ip interface brief

```
Router#show ip interface brief
Interface                IP-Address      OK? Method Status      Protocol
GigabitEthernet0/0       10.0.1.1        YES manual up          up
GigabitEthernet0/1       10.0.3.1        YES manual up          up
GigabitEthernet0/2       10.0.2.1        YES manual up          up
Vlan1                    unassigned      YES unset  administratively down down
```

Task 55

```
C:\>ping 10.0.2.10

Pinging 10.0.2.10 with 32 bytes of data:

Request timed out.
Request timed out.
Request timed out.
Request timed out.

Ping statistics for 10.0.2.10:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss)

C:\>tracert 10.0.2.10

Tracing route to 10.0.2.10 over a maximum of 30 hops:

  0  *         *         *         Request timed out.
  1  *         *         *         Request timed out.
  2  *         *         *         Request timed out.
  3  *         *         *         Request timed out.
```

Task 56

```
C:\>ping 10.1.2.10

Pinging 10.1.2.10 with 32 bytes of data:

Request timed out.
Request timed out.
Request timed out.
Request timed out.

Ping statistics for 10.1.2.10:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),
```

Task 57

R2

```
Router(config)#interface GigabitEthernet0/0
Router(config-if)#ip address 10.0.0.1 255.255.255.0
Router(config-if)#no shutdown
Router(config-if)#int g0/1
Router(config-if)#ip address 10.1.0.0 255.255.255.0
Bad mask /24 for address 10.1.0.0
Router(config-if)#no shutdown
```

R3

```
Router(config-if)#int g0/1
Router(config-if)#ip address 10.1.1.1 255.255.255.0
% 10.1.1.0 overlaps with GigabitEthernet0/0
Router(config-if)#no shutdown
Router(config-if)#int g0/1
Router(config-if)#ip address 10.2.1.1 255.255.255.0
Router(config-if)#no shutdown
```

R4

```
Router(config)#interface GigabitEthernet0/0
Router(config-if)#ip address 10.1.1.0 255.0.0.0
```

Task 61

Pc3 ping from Pc1

```
Pinging 10.1.2.10 with 32 bytes of data:

Request timed out.
Request timed out.
Request timed out.
Request timed out.

Ping statistics for 10.1.2.10:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),
```

Task 62

```
Router>ena
Router#conf t
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#ip route 10.1.0.0 255.255.0.0 10.0.0.2
Router(config)#exit
```

Task 63

```
Router#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

10.0.0.0/8 is variably subnetted, 6 subnets, 2 masks
C       10.0.1.0/24 is directly connected, GigabitEthernet0/0
L       10.0.1.1/32 is directly connected, GigabitEthernet0/0
C       10.0.2.0/24 is directly connected, GigabitEthernet0/2
L       10.0.2.1/32 is directly connected, GigabitEthernet0/2
C       10.0.3.0/24 is directly connected, GigabitEthernet0/1
L       10.0.3.1/32 is directly connected, GigabitEthernet0/1
```

Task 66

```
Router>ena
Router#conf t
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#ip route 10.1.3.0 255.255.255.0 10.0.3.2
Router(config)#end
```

Task 67

```
C:\>ping 10.1.3.2

Pinging 10.1.3.2 with 32 bytes of data:

Request timed out.
Request timed out.
Request timed out.
Request timed out.

Ping statistics for 10.1.3.2:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),
```

Task 68

```
Router>ena
Router#conf t
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#int g0/1
Router(config-if)#ip address 203.0.113.1 255.255.255.0
Router(config-if)#no shutdown
```

Task 69

```
Router>ena
Router#conf t
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#ip route 0.0.0.0 0.0.0.0 10.0.0.2
Router(config)#end
```

Task 70

```
Router#conf t
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#ip route 0.0.0.0 0.0.0.0 10.0.0.2
Router(config)#ip route 0.0.0.0 0.0.0.0 10.0.3.2
Router(config)#end
```

- 71. Yes. R1 automatically adds connected routes for its directly connected networks.
- 72. PC1 loses reachability because R1 no longer has a route to PC3's network.
- 73. A summary route combines multiple smaller networks into one larger network prefix, so one route can represent several /24 networks.